

① What is data science Explain its primary objective Date: 10/07/20

Data science is an interdisciplinary field that uses scientific methods, algorithms and statistical techniques to extract meaningful insights from data. It combines mathematics, programming and domain knowledge to analyze data.

The primary objective of data science is to convert raw data into useful information for better decision making.

② Why is data science important in today's digital world?

- Helps organization make data-driven decisions.
- Predicts future trends using historical data.
- Improves customer experience through personalization.
- Increases efficiency by automating data analysis and business processes.

③ List any four benefits of data science in business and society?

- Better business decision making.
- Early detection of fraud and cyber threats.
- Improved healthcare through disease prediction.
- Efficient resource planning and smart city development.

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4. Mention four real world applications of data science.

- Healthcare - Disease diagnosis and Patient monitoring
- Banking - Fraud detection and credit risk analysis
- E-commerce - Product recommendation and customer analytics
- Transportation: Traffic Prediction and route optimization.

5. What are the different facets of data?

- structured data: organized in rows and columns
- semi structured data: Has a flexible structure using tag or keys.
- unstructured data: Has no predefined format.

6. Diff between structured, semi structured and unstructured data with suitable examples

Structured data: organized in row and column and easy to store and query
Ex: SQL Database, Excel sheet.

semi-structured data: Has a flexible structure with tag or keys and partially organized. Ex: XML, JSON.

Unstructured data: Has no predefined format and difficult to process directly. Ex: Images, videos, emails, social media posts.

7. What is Big data state its key characteristics?

Big data refers to extremely large and complex datasets that cannot be processed using traditional data processing tools. Its key characteristics are known as the 5 Vs.

- Volume
- velocity
- Variety
- veracity
- Value

8. Explain the Big data ecosystem?

The big data ecosystem consists of tools and technologies used to collect, store, process, analyze, and visualize large datasets. It enables efficient handling of massive volumes of structured and unstructured data.

-> Technologies: Hadoop, Apache Spark, Hive, HBase.

9. What are the major phases of the data science process?

(i) Data collection -> data cleaning -> Data integration and transformation -> Data Analysis and Model Building -> Model Evaluation -> Deployment and Result communication.

10. What is data retrieval? Mention two common sources from which data can be retrieved?

Data retrieval is the process of obtaining required data from storage systems for analysis or processing. It ensures that relevant information is available when needed.

Common data sources: Database (SQL), APIs or web services

Why is data collection considered an important step in data science process?

Data collection is the first step in the data science process as the quality of data directly affects the accuracy of results. It gathers relevant information from various sources for analysis.

12. what is data cleansing? List any four common data cleansing techniques in data science.
Data cleansing is the process of identifying and correcting errors, inconsistencies and missing values in a data set to improve its quality.
Common techniques: Handling missing values, Removing duplicate records, correcting inconsistent data, detecting and removing outliers.

13. what is data integration why it's necessary in data science?
Data integration is the process of combining data from multiple sources into a single, unified data set. It ensures consistency and eliminates redundancy. It is necessary because it provides a complete view of the data, improving analysis and decision making.

14. Explain data transformation with suitable example.

Data transformation is the process of converting data into suitable format for analysis or model building.
It improves data quality and compatibility.

Example:

- Converting categorical values into numerical values
- Normalizing or scaling numerical data
- Changing date formats.

⑥ what is data analysis ?

Data analysis is the process of examining data to discover useful patterns, trends and insights for decision making.

Techniques:-

- Descriptive Analysis
- Diagnostic Analysis
- Predictive Analysis
- Prescriptive Analysis

⑦ Diff between descriptive, predictive and prescriptive analytics

→ Descriptive Analytics:- It explains what happened and uses historical data.

→ Predictive Analytics:- Predicts what may happen and uses statistical and ML models.

→ Prescriptive Analytics:- suggests what action to take and uses optimization and AI.

⑧ What is meant by building a model is data science.

Model building is the process of creating a machine learning or statistical model to identify patterns and makes predictions from data. select the algorithm →

- Train the model using train data
- Test the model using test data
- improves the model by tuning its parameters

19) Why is presenting finding an essential part of the data science process?

Presenting finding helps stakeholders understand the insights obtained from data and support informed decision making. Effective visualization makes complex data easy to interpret.

20) How are data science models deployed into real world applications?

After evaluation, a data science model is deployed into real world systems using web applications, cloud platforms, or APIs so users can access predictions in real time.

Applications:

- Fraud detection
- Recommendation systems
- Disease Prediction.
- Spam Email detection.